Learning Verified Monitors for Hidden Markov Models

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Abstract. Runtime monitors for hidden Markov models obtain a sequence of observations and must classify whether the system is safe or unsafe. That is, the probability of being in a dangerous state conditioned on these observations exceeds some acceptable threshold. The key contribution of this paper is the first correct-by-construction synthesis method for such monitors, represented as finite state machines. First, we establish the coNP-hardness of checking whether a finite state machine is indeed a correct monitor, i.e., a monitor without misclassifications. Second, we provide a reduction that reformulates the search for misclassifications into a standard probabilistic system synthesis problem. Third, we integrate the verification routine into an active automata learning routine to synthesise correct monitors. Fourth, we provide a prototypical implementation that shows the feasibility and limitations of the approach on a series of benchmarks.

1 Introduction

Runtime assurance is an essential ingredient in the deployment of safe autonomous systems. Runtime monitors provide assurance by flagging potentially dangerous system behavior by observing a system execution. More precisely, a monitor receives a stream of observations about the system and outputs a verdict, e.g., it raises an alarm that the system has left some safety envelope. A monitor is correct, if it correctly raises such alarms based on a formal specification. Various challenges in creating correct runtime monitors for (semi-)autonomous systems have been identified [**empty citation**], such as: (1) the state of the system is only partially observable, i.e., the stream of observations comes from sensor readings and does uniquely identify the state of a system, (2) the behavior of the system and/or the sensors may be subject to probabilistic uncertainty, (3) the monitor itself is subject to resource constraints (time, memory, etc), and (4) the monitor is itself safety-critical and should therefore be subject to extensive validation. Challenges (1,2) can be (partially) addressed by modelling the system as a hidden Markov model (HMM), Challenges (3,4) can be (partially) addressed by representing a monitor as, e.g., a (small) finite state machine. Concretely, this paper focuses on the following question: *Is a given finite state machine a correct monitor for a given HMM?* The paper studies the complexity of this problem, provides a practical verification procedure, and embeds this verification procedure into a learning framework.

What are HMMs? In this paper, we assume that the system including its sensors is adequately described as a discrete Hidden Markov Model (HMM). Markov chains (MCs) describe system behavior subject to probabilistic uncertainty. Paths through an MC are sequences of states that describe system executions. HMMs extend MCs by labelling their states with colours (aka *observations*). Intuitively, the observations can be used to model the information that the monitor receives. In HMMs, every path can be lifted to a sequence of observations, which we call a *trace*. The trace associated to a system execution describes the information received by the monitor.

Risk of a trace. In runtime monitoring, an essential task is to estimate whether the current system is dangerous. In the simplest setting, a specification describes a set of states is dangerous. The risk of a trace is then the probability of ending in a dangerous state, conditioned on the fact that the system execution matches the trace. For a given trace, we may compute this risk, e.g., either via a reduction to model checking [**empty citation**] or by a (forward) filtering that tracks a distribution over the current states [**empty citation**].

What are correct monitors? Often, we are not interested in this precise risk, but rather want to partition the set of horizon-bounded traces into safe and unsafe traces: Unsafe traces are traces whose risk does exceed some fixed *acceptable threshold*, all other traces are safe. Indeed, a *correct* monitor obtains a trace and raises an alarm iff a trace is not safe. The key verification problem in this paper asks whether a monitor, represented as a deterministic finite automaton, accepts (i.e., raises an alarm) exactly the unsafe traces. We note that in this paper, we only consider traces that are bounded by some fixed horizon.

Illustrative toy example. SJ: TODO make more concrete, add Fig. As an running example, we consider an oversimplified car driving scenario, loosely inspired by runtime monitors obtained from high-fidelity simulations [**empty citation**]. A car can be in three states: It can be on a dry road, on an icy patch, or it can drifted off the road. The HMM in Figure 5 describes the situation, where a car alternates between dry and icy road segments, and where being on a icy (dry) segment positively affects the probability that the next segment is icy (dry). When on an icy road, there is a higher probability to drift off the road. Our sensor can detect dry roads, but cannot distinguish between icy roads and off-road conditions. We can define the trace risk as the probability to be off-road, conditioned on a a trace, such as $dry \cdot ice \cdot ice$. In **??**, we depict a finite automaton that describes the set of traces that end with two consecutive icy patches. The key verification question in this paper is now: Is this a correct monitor for threshold 0.9? Put differently, does it raise an alarm iff the trace risk is above 0.9.

SJ: I parked this figure here from the proposal

SJ: Do we also want a small figure of a system feeding observations into a monitor?

Novel complexity results. Deciding whether a monitor is correct is coNP-complete (Theorem 2). In particular, we study the dual problem that asks whether a monitor misclassifies at least one trace. We consider a simpler variant of this problem that asks whether there is a bounded trace in the HMM that is unsafe.

[1] The problem is clearly in NP (assuming that the bound is unary encoded). We show that this simplified problem is also strongly NP-hard (Lemma 4) and the corresponding optimization problem, that asks to approximate the risk of the trace with the heighest risk, is APX-hard (Lemma 5). Intuitively, APX-hardness means that the existence of an efficient approximation algorithm would have major complexity-theoretic implications.

Our verification approach. While deciding whether a monitor is correct is thus in general not tractable, we suggest to utilise recent advances in the synthesis of probabilistic systems. We concentrate on proving the absence of missed alarms, i.e., we concentrate on showing that the monitor correctly identifies every unsafe trace. A similar reduction works for showing the absence of spurious alarm. First, for a single trace, computing the risk can be reduced to computing a reachability probability in a MC that is intuitively a kind of product between a DFA that accepts exactly the trace and the HMM [**empty citation**]. Thus, it is natural to reduce our problem to the question: Is there a DFA (accepting exactly one trace, accepted by our monitor) such that the probability in the product MC exceeds a threshold. If the answer is no, then the monitor has no missed alarms. This decision problem can be solved using (exact) probabilistic model synthesis methods [**empty citation**] as implemented in the tool Paynt [**empty citation**][2]. We discuss our verification approach in Section 3.

Our learning approach. Ultimately, we do not only want to verify the correctness of a monitor, but we want to construct such a monitor. Above, we already mentioned that the monitors are formal languages. When considering bounded trace, these monitors are regular, and can thus be captured using DFAs. Inspired by a recent short paper [**empty citation**], we investigate learning these monitors: We choose to do this using active automata learning [3, 8], which closely relates to other popular oracle-guided inductive synthesis loops [**empty citation**]. To use active automata learning (AAL), we must provide a *membership oracle* that

[1] The results also apply to checking for safe traces.

[2] Alternatively, in Section **??**, we show that we can construct a product of an acyclic DFA accepting sets of traces and the HMM. However, this yields partially observable MDP, where we must search for memoryless policies, similar to a construction in [4]. This problem is equivalent to the synthesis problem and can therefore also be solved using Paynt [1].

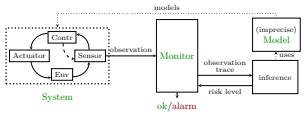


Fig. 1: White-box monitoring of systems

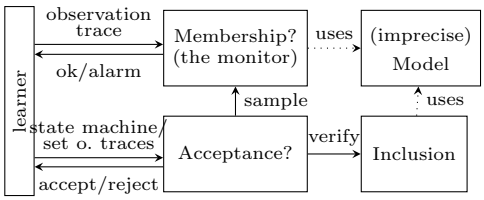
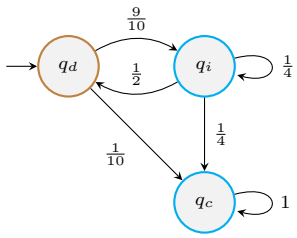
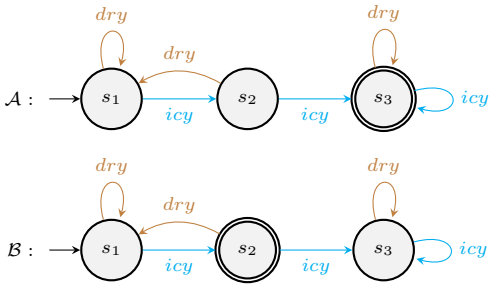


Fig. 2: Learning monitors.



(a) Example HMM, where the cyan states q_i and q_c have the observation *icy* and the brown state q_d has the observation *dry*.



(b) Two example monitors for the HMM in fig. 3a, correct monitor $\mathcal{A}$ and faulty monitor $\mathcal{B}$. $\mathcal{A}$ raises an alarm as soon as two steps on an *icy* section have been observed. While $\mathcal{B}$ raises an alarm after exactly one step on icy road has been observed.

Fig. 3: An HMM and a correct monitor that raises an alarm if the probability of being off-road is above 0.8.

decides for a single trace whether the risk of the trace exceeds a threshold and an *equivalence oracle* that decides whether a given hypothesis monitor is indeed correct. The membership oracle can be implemented by computing the risk for a single trace as above and comparing it to a threshold, while the equivalence oracle can be implemented by checking for the absence of missed and false alarms. Practically, some minor modifications are necessary to embed our approach in AAL framework: We have to handle the finite horizon, resolve some traces that are not relevant for the monitor, and we combine conformance and equivalence queries to boost performance.

Putting it together. In Fig. 4, we outline our proposed framework to learn a monitor. SJ: TODO explain framework

Empirical evaluation. SJ: TODO summarise results of the empirical eval here.

Contributions. In summary, this paper provides a first framework to learn runtime monitors — encoded as finite automata — that are verified to be correct with respect to a given HMM. The main contributions are as follows: (1) We study monitor verification and show complexity-theoretic hardness of the underlying problem, even in an approximative setting. (2) We reduce the monitor verification to probabilistic model synthesis problems that exhaustively search for a counterexample. (3) We learn monitors by using the verifier above to answer equivalence queries in conjunction with active automata learning. (4) We demonstrate the feasibility and limitations of both the verifier and the learner on a series of benchmarks. Our prototype can handle HMMs with thousands of states, up to a hundred observations, and monitors with over 100 states.

TODO add our framework

Fig. 4

2 Problem Statement

In this section, we recap standard notions on stochastic models and finite automata and introduce the formal problem statement.

Notation. Let X be a finite set. A distribution μ over X is a mapping $\mu\colon X \to [0,1]$ such that $\sum_{x \in X} \mu(x) = 1$. The set of all distributions over X is written $\Delta(X)$.

2.1 Models

In this section, we briefly introduce MDPs, which are important later, and HMMs which are crucial to the problem statement. Details can be found in [5].

Definition 1 (MDP). *A* Markov decision process *(MDP) is a tuple* $\mathcal{M} = (S, \iota, Act, \mathbf{P})$ *with a countable nonempty set* S *of* states*, the* initial state $\iota \in S$*, the* partial transition function $\mathbf{P}\colon S \times Act \rightharpoonup \Delta(S)$*.*

We use $Act(s) = \{a \mid \mathbf{P}(s,a) \neq \bot\}$ as the set of enabled actions. We assume that there are no deadlocks, i.e., for every $s \in S$, $Act(s) \neq \emptyset$. A path π in an MDP is a sequence $s_0 a_0 s_1 \ldots \in (S \times Act)^* \times S$, such that $a_i \in Act(s_i)$ and $P(s_i, a_i)(s_{i+1}) > 0$ for every $i \geq 0$. The set of (finite) paths is denoted by $\Pi^{\mathcal{M}}_{fin}$, the set of paths of at most length h are $\Pi^{\mathcal{M}}_{h}$, and the set of paths of exactly length h are $\Pi^{\mathcal{M}}_{=h}$. If the MDP is clear from the context, the $\mathcal{M}$ is dropped.

A policy for an MDP is a function $\sigma\colon \Pi^{\mathcal{M}} \to Act$, which given a path through the MDP, gives a action. Given a policy, a MDP can be induced into a MC. Policy synthesis for an MDP for a property ϕ thus entails finding a policy for an MDP such that the induced MC entails ϕ.

A *Markov chain* (MC) is an MDP where $|Act(s)| = 1$ for every state $s \in S$. We simplify notation and write MCs also as tuple $(S, \iota, \mathbf{P})$ and write $\mathbf{P}(s)$ to refer to the unique distribution $\mathbf{P}(s,a)$. Paths in a MC can be simplified to be sequences of (only) states.

The probability measure $Pr^{\mathcal{M}}$ of an MC $\mathcal{M}$ is the unique probability measure following from the canonical σ-algebra associate with *mc*.

Luko: Condition probability on traces defined using Paths

Definition 2 (HMMs). *A (risk-labelled) HMM is a tuple* $(S, \iota, \mathbf{P}, Z, obs, r)$ *such that* $(S, \iota, \mathbf{P})$ *is a MC,* Z *is a finite set of observations, and* $obs\colon S \to Z$ *is the (deterministic) observation function*[3]*, and* $r\colon S \to \mathbb{R}_{\geq 0}$*,*

[3] We use determinstic observation functions for concise definitions. Non-deterministic observation functions can be expressed via a blowup of the HMM, see e.g. [**empty citation**]

Notions such as paths are lifted from MCs. Furthermore, traces τ are sequences of observations, and the set of traces of a HMM is the language $\mathcal{L}(\mathcal{M})$ of the HMM. The set $\text{Paths}(\tau)$ contains all paths that correspond to a trace τ, $\text{Paths}(\tau) = \{\pi \in \Pi^{fin} \mid obs(\pi) = \tau\}$. SJ: Also needs things like the probability of a trace.

Example 1.

Definition 3. *A deterministic finite automaton (DFA) is a tuple $\mathcal{A} = (Q, \Sigma, \delta, \iota, F)$. Q is a finite set of states, Σ is an alphabet, $\delta \colon Q \times \Sigma \rightharpoonup Q$ is a (partial) transition function, ι is the initial state, and $F \subseteq Q$ is a set of accept (or: final) states.*

Let ε denote the empty word. We lift the transition relation to words, which is defined recursively: $\delta^*(q, \varepsilon) = \varepsilon$ if $a \in \Sigma$ and $\delta^*(q, w \cdot a) = \delta(\delta^*(q, w), a)$.

The *language* $\mathcal{L}(\mathcal{A})$ of a DFA $\mathcal{A}$ is the set of all words accepted by the $\mathcal{A}$, $\mathcal{L}(\mathcal{A}) = \{w \in \Sigma^* \mid \delta^*(\iota, w) \in F\}$.

2.2 Formal Problem Statement

The monitor verification problem consists of two sub problems. The no false alarms problem, and the no missed alarms problem. Both of these sub problems entail finding the trace with either the minimum or maximum risk. We will discuss the missed alarm problem in the following sections. In section 3.4 we will adapt to the no false alarms problem.

The no missed alarms problem entails ensuring that all traces from the MC, which are not in the monitor, have at most a risk of λ.

Definition 4 (Risk of a trace). *Given a HMM $\mathcal{M}$. The risk of trace $\tau \in \mathcal{L}(\mathcal{M})$ is:*

$$R_r(\tau) := \sum_{\pi \in \Pi^h_{\mathcal{M}}} Pr(\pi \mid \tau) \cdot r(\pi_{\downarrow}).$$

The risk of a trace can be used to partition the set of all traces.

Definition 5 (Safe traces/Unsafe traces). *Given a HMM $\mathcal{M}$, a* safe threshold *$\lambda_s \in \mathbb{R}$ and an* unsafe threshold *$\lambda_u \in \mathbb{R}$ with $\lambda_s \leq \lambda_u$. A trace $\tau \in \mathcal{L}(\mathcal{M})$ with $R(\tau) > \lambda_u$ is an* unsafe trace *(for $\mathcal{M}$ and λ_u). Conversely, trace τ is a* safe trace*, if $R(\tau) < \lambda_s$ (for $\mathcal{M}$ and λ_u).*

In the definition above, we deliberately do not require $\lambda_s = \lambda_u$. In particular, by picking $\lambda_s < \lambda_u$, some traces are neither safe nor unsafe. We write $\mathbb{S}^{\leq h}_{\mathcal{M},\lambda_s}$ to denote the set of safe traces of length at most h and $\mathbb{U}^{\leq h}_{\mathcal{M},\lambda_u}$ for the set of unsafe traces of length at most h. We often omit the sub/superscripts.

Example 2.

We are now ready to define correct monitors.

Fig. 5: Central example to the paper

Definition 6 (Monitor). *A DFA $\mathcal{A}$ is a monitor for HMM $\mathcal{M}$ if the alphabet for $\mathcal{A}$ coincides with the observations in $\mathcal{M}$.*

Definition 7 (Missed/False alarms). *Given a monitor $\mathcal{A}$ for HMM $\mathcal{M}$, a horizon h, and thresholds $\lambda_s \leq \lambda_u \in \mathbb{R}$, the set of* missed alarms *is* $\mathsf{mA}^{\leq h}_{\mathcal{M},\lambda_s}(\mathcal{A}) := \mathbb{U}^{\leq h}_{\mathcal{M},\lambda_u} \setminus \mathcal{L}(\mathcal{A})$. *The set of* false alarms *is* $\mathsf{fA}^{\leq h}_{\mathcal{M},\lambda_s}(\mathcal{A}) := \mathbb{S}^{\leq h}_{\mathcal{M},\lambda_s} \cap \mathcal{L}(\mathcal{A})$.

We often drop the sub/superscripts. In terms of a classification problem, missed alarms are false negatives, while false alarms are false positives.

Definition 8 (Correct monitor). *Given thresholds λ_s, λ_u and horizon h, a monitor $\mathcal{A}$ for $\mathcal{M}$ is* correct *if* $\mathsf{mA}^{\leq h}_{\mathcal{M},\lambda_s}(\mathcal{A}) = \emptyset = \mathsf{fA}^{\leq h}_{\mathcal{M},\lambda_u}(\mathcal{A})$.

A correct monitor raises an alarm for all unsafe traces and for no safe trace:

Corollary 1. *A monitor $\mathcal{A}$ is correct iff* $\mathbb{U}^{\leq h}_{\mathcal{M},\lambda_u} \subseteq \mathcal{L}(\mathcal{A}) \subseteq \Sigma^* \setminus \mathbb{S}^{\leq h}_{\mathcal{M},\lambda_s}$.

Problem statements. This paper tackles the following problem statements: Given HMM $\mathcal{M}$, thresholds λ_s, λ_u and horizon h,

1. Given monitor $\mathcal{A}$ for $\mathcal{M}$, are there missed alarms, i.e., is $\mathsf{mA}^{\leq h}_{\mathcal{M},\lambda_s}(\mathcal{A}) = \emptyset$?
2. Given monitor $\mathcal{A}$ for $\mathcal{M}$, are there false alarms, i.e., is $\mathsf{fA}^{\leq h}_{\mathcal{M},\lambda_u}(\mathcal{A}) = \emptyset$?
3. Find a correct monitor $\mathcal{A}$ for $\mathcal{M}$ w.r.t. λ_s, λ_u and h.

Problem statements 1 and 2 together allow checking whether a monitor is correct.

Remark 1. A correct monitor must exist, as $\mathbb{U}^{\leq h}$ is finite and thus regular.

We discuss our solution to the no missed alarms problem in section 3, we adapt the solution to the no false alarm problem in section 3.4.

Central example We now introduce the central example used in this paper. And explain how the monitor verification problem and verified monitor learning problem apply to it.

Example 3. Given the risk function assigning 1 to q_c and 0 to all other states, the traces $\tau_1 = dry, icy, icy$ and $\tau_2 = dry, icy$ have risks $\frac{19}{40}, \frac{4}{40}$ respectively. If λ is $\frac{1}{4}$ and the horizon h is 3, τ_2 is the trace with maximum risk not in the monitor. And its risk is below λ. Thus the monitor does not have an missed alarms.

Luko: Include verified monitor learning problem

3 Monitor Verification

In this section we present our approach to the monitor correctness problem, which essentially reduces checking the existence of missed alarms to a policy synthesis problem on colored MDPs. The latter is well studied. We first formalise this policy synthesi problem and then discuss the transformation in multiple steps. In our presentation, we focus on the no missed alarms problem with traces of exactly the length of the horizon. At the end of the section, we generalize our construction to both the no false alarms problem and traces of length at most the horizon.

3.1 Relating Missed Alarms with Color-Consistent Policies

Colored (Labelled) MDPs are a mild extension to MDPs that allow expressing dependencies between states that policies must adhere to. The following definition simplified the notion of colored MDPs from [**empty citation**] suffices for our needs:

Definition 9 (Colored MDP). *Given an MDP $\mathcal{M}$ with states S, a* colored MDP *is a tuple $\mathcal{M}^C = (\mathcal{M}, C, c)$, where C is a set of* colors*, and $c\colon S \to C$ maps states to colors.*

Definition 10 (Color-consistent scheduler). *A memoryless scheduler σ for a colored MDP $\mathcal{M}^C$ is* color consistent[4] *if for every two states s and s'*

$$c(s) = c(s') \quad \textit{implies} \quad \sigma(s) = \sigma(s').$$

The set of all color-consistent schedulers is denoted Σ_c. Policy synthesis for colored MDPs asks to find a color consistent policy such that the induced reachability probability is above (or below) a certain threshold. Policy synthesis for colored MDPs is NP-hard.

Theorem 1 (Main theorem). *Given a HMM $\mathcal{M}$ with a monitor $\mathcal{A}$, threshold λ_s, horizon h, and risk r, there exists a colored MDP $\mathcal{M}^C$ and a λ such that*

$$\exists \sigma \in \Sigma_c . Pr_\sigma^{\mathcal{M}^C}(\Diamond T) \geq \lambda \quad \textit{iff} \quad \exists \tau \in \mathsf{mA}^{=h}_{\mathcal{M},\lambda_s}(\mathcal{A}).$$

Luko: Should we also list the risk function r in this theorem?

Our proof is constructive and we show that we can use the construction to find a $\tau \in \mathsf{mA}(\mathcal{A})$, whenever such a τ exists.

Outline of the proof. The proof is a direct consequences of Lemmas 1 and 2 and **??** below. Before we discuss these lemmas, we observe that on the left-hand side of Theorem 1, the monitor, horizon, observations, and risk do not occur, they must be encoded into the colored MDP. Furthermore, while missed alarms are defined using conditional probabilities, the policy synthesis problem is over

[4] Colored MDPs with color-consistent policies coincide with memoryless policies for partially observable MDPs. However, POMDPs often consider history-dependent (belief-based) policies. We use colored MDPs to avoid any confusion.

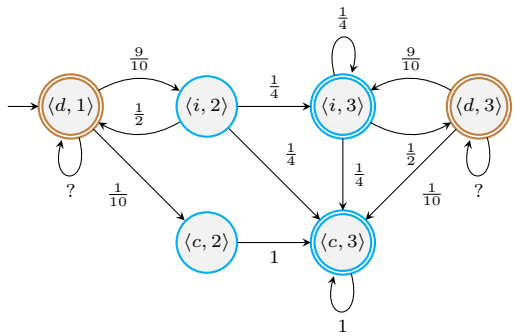


Fig. 6: The central example HMM and faulty monitor $\mathcal{B}$ transformed into the CTP problem. States are named with their HMM state, $\{d, i, c\}$ and their monitor state, $\{1, 2, 3\}$. States in the alarm states set A have been marked accepting.

reachability probabilities. We describe our transformation in several steps. In Section 3.2 we encode the monitor into the HMM and transform the HMM to both include the horizon and the risk. In Section 3.3 we resolve the conditioning on traces, while replacing observations from the HMM.

3.2 The (Acyclic) Conditional Trace Probability Problem

First, we show how asking for a missed alarm can be rephrased into the conceptually simpler *conditional trace probability* (CTP) problem. We will further simplify the problem by incorporating both the horizon and risk such that we are left with a problem purely on acyclic HMMs.

CTP problem Before we combine the HMM and the monitor, we have to modify the monitor. Since we are looking for missed alarms, we are interested in all traces that are not in the monitor. We thus take the complement of the monitor. Next, we take (a mild variation of) a standard product construction [5, Section 10.3] between HMM and DFA. We also keep track of all resulting states for which the complement monitor accepts in a set of alarms states A of the product HMM.

Example 4. Figure 6 shows the product of the HMM and monitor $\mathcal{B}$ from Figure 5. In particular, when starting in the initial states d and 1, respectively, the HMM can transition to the i state with probability $\frac{9}{10}$. This is an *icy* state, and thus the monitor takes the *icy* transition to the 2 state. This results in the new product state of i and 2, with a transition from $\langle d, 1\rangle$ to $\langle i, 2\rangle$ with probability $\frac{9}{10}$.

The set of Alarm states are all the states in monitor $\mathcal{B}$ which do not accept. Thus they are the initial $\langle d, 1\rangle$ state and any product states where the monitor is in state 3.

Definition 11 (HMM product). *Given a HMM $\mathcal{M} = (S, \iota^{\mathcal{M}}, \mathbf{P}, Z, obs, r)$ and a monitor for $\mathcal{M}$, $\mathcal{A} = (Q, \Sigma, \delta, \iota^{\mathcal{A}}, F)$. The product HMM $\mathcal{M}_{\times\mathcal{A}}$ is the HMM:*

$$\begin{aligned}
\mathcal{M}_{\times\mathcal{A}} &= (S \times Q, \langle \iota^{\mathcal{M}}, \iota^{\mathcal{A}} \rangle, \mathbf{P}', obs', r') \\
\mathbf{P}'(\langle s, q \rangle)(\langle s', q' \rangle) &= \begin{cases} \mathbf{P}(s, s') & \text{if } \delta(q, obs(s')) = q' \\ 0 & \text{otherwise} \end{cases} \\
obs'(\langle s, q \rangle) &= obs(s) \\
r'(\langle s, q \rangle) &= r(s)
\end{aligned}$$

The alarm states become:

$$A = S \times F$$

In the product, we can find a trace whose conditional trace probability exceeds a threshold iff there is a missing alarm. We state the decision problem that needs to be solved on the product HMM as it is key to our computational complexity analysis in Section 5.

Definition 12 (CTP Decision Problem). *Given a HMM $\mathcal{M}$ with states in S, a unary encoded horizon h, a set of alarm states $A \subseteq S$, and a threshold $\lambda_s \in (0, 1]$,*

$$\exists \tau \in \mathcal{L}(\mathcal{M}). \sum_{\pi \in \Pi^{\mathcal{M}}_{=h} \mid \pi_{\downarrow} \in A} Pr^{\mathcal{M}}(\pi \mid \tau) \cdot r(\pi_{\downarrow}) \geq \lambda.$$

We denote the set of witnesses τ to a CTP decision problem as $\mathsf{CTP}(\mathcal{M}, h, A, \lambda_s)$. The following lemma states the correctness of the transformation and follows directly from the definition of missing alarms and the product with the complement.

SJ: Nevetheless, we should have a (slightly) more formal proof

Lemma 1. *Using the notation from Theorem 1, Definition 11 and Definition 12:*

$$\exists \tau \in \mathsf{CTP}(\mathcal{M}_{\times\overline{\mathcal{A}}}, h, A, \lambda_s) \quad \textit{iff} \quad \exists \tau \in \mathsf{mA}^{=h}_{\mathcal{M}, \lambda_s}(\mathcal{A}).$$

ACTP problem We further simplify the problem by combining the horizon, risk and the newly created alarm states set into the HMM. The resulting HMM can be seen as the model for the acyclic CTP problem.

Example 5. The ACTP transformation applied to the HMM from fig. 6 can be seen in fig. 7. We unroll the CTP HMM for a horizon of 3. Thus, the initial state becomes the tuple of step 1 and the initial state from the CTP HMM, $\langle d, 1 \rangle$. Any transition in the model increases the step value by 1, $\langle 1, \langle d, 1 \rangle \rangle$ transitions to $\langle 2, \langle c, 2 \rangle \rangle$. In state $\langle 3, \langle i, 3 \rangle \rangle$ we are at the horizon, and the CTP part of the state is part of the alarm states. Thus, the monitor wants to raise an alarm here. We now transition with probability 1 to the $\langle 4, safe \rangle$ state and probability 0 to the $\langle 4, alarm \rangle$ state, since the normalized risk of d is 0. The horizon state $\langle 3, \langle d, 1 \rangle \rangle$ is not in A, thus we transition to the $\langle 4, ignore \rangle$ state.

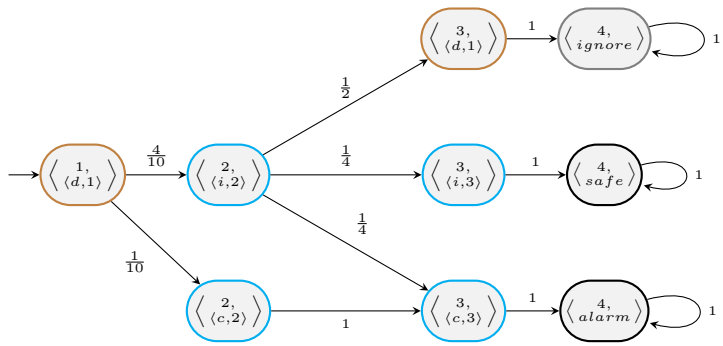


Fig. 7: ACTP, brown and cyan are dry and icy observations. Black is the new end observation, and gray is the new ignore observation. All states are named with the step i, model state s and monitor state j as $\left\langle \begin{smallmatrix} i, \\ \langle s,j \rangle \end{smallmatrix} \right\rangle$.

Definition 13 (Unrolling). *The (risk-)unrolled HMM $\mathcal{M}_{\blacktriangleright h}$ of a HMM $\mathcal{M} = (S, \iota, \mathbf{P}, Z, obs, r)$ with horizon h, risk r and alarm states $A \subseteq S$ is*

$$\mathcal{M}_{\blacktriangleright h} = \left(\{1, \ldots, h\} \times S \cup \{alarm, safe, ignore\}, (1, \iota), \mathbf{P}', Z \cup \{end, ignore\}, obs'\right.$$

The risk function is normalied as r':

$$r'(s) = \frac{r(s)}{\max_{s \in S} r(s)}$$

The unrolled transition relation $\mathbf{P}'$ is defined as expected. At the horizon we transition to the three new states alarm, safe and ignore[5] *according to the risk.*

$$\forall_{i \in \{1,\ldots,h-1\}} \quad \mathbf{P}'(\langle i, s\rangle)(\langle i+1, q\rangle) = \mathbf{P}(s)(q)$$

$$\mathbf{P}'(\langle h, s\rangle)\,(\langle h+1, a\rangle) = \begin{cases} r'(s) & \text{if } s \in A \wedge a = alarm \\ 1 - r'(s) & \text{if } s \in A \wedge a = safe \\ 1 & \text{if } s \notin A \wedge a = ignore \\ 0 & \text{otherwise} \end{cases}$$

$$\mathbf{P}'\left(\langle h+1, a\rangle\right)\left(\langle h+1, a\rangle\right) = 1$$

We observe the end and ignore observations in the $\langle h+1, alarm\rangle$, $\langle h+1, safe\rangle$ and $\langle h+1, ignore\rangle$ states respectively.

$$obs'(\langle i, s\rangle) = obs(s)$$

$$obs'(\langle h, a\rangle) = \begin{cases} end & \text{if } a \in \{alarm, safe\} \\ ignore & \text{if } a = ignore \end{cases}$$

Remark 2. Note that we ignore all states in A that are not horizon states. However, since we only consider traces which are of length exactly the horizon, we only need to consider the alarm states which are encountered after horizon steps.

SJ: not sure what the remark states

Luko: There are also states in A which are not on the horizon. They are not conisdered since they do not produce traces of length h.

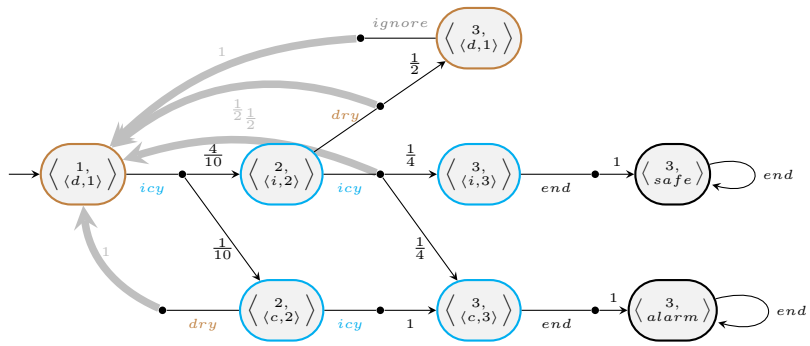


Fig. 8: The central example transformed into an mdp to be used as the model in the MDP policy synthesis problem. Every state has an action which contains only transitions to states which had the observation related to that action.

bad explanation

Lemma 2. *Given an HMM $\mathcal{M}$, horizon h, alarm states A and threshold λ:*

$$\exists \tau \in \mathsf{CTP}(\mathcal{M}, h, A, \lambda_s) \quad \textit{iff} \quad \exists \tau \in \mathcal{L}(\mathcal{M}_{\blacktriangleright h}). \sum_{\pi \in \Pi^{\mathcal{M}}} Pr^{\mathcal{M}}(\pi s_a \mid \tau \cdot z_{\mathsf{end}}) \geq \lambda,$$

where z_{end} is the final observation and s_a is the alarm state.

SJ: I wonder whether we should simplify notation on the right hand side. I am not too fond of the z and s_a symbols.

The proof of the lemma essentially starts by using that the lhs of the lemma is by definition $\exists \tau \in \mathcal{L}(\mathcal{M}).\ \sum_{\pi \in \Pi^{\mathcal{M}}_{=h} \mid \pi_{\downarrow} \in A} Pr^{\mathcal{M}}(\pi \mid \tau) \cdot r(\pi_{\downarrow}) \geq \lambda$.

SJ: keep this in sync with def definition 12

3.3 Reduction to Consistent Policy Synthesis

For Lemma 2, we must find a trace such that a conditional reachability probability exceeds a threshold. We reformulate this into a policy synthesis problem in MDPs. The transformation combines two ideas: First, in every state, the policy can select the next observation, inspired by [**empty citation**]. Second, we reformulate conditional reachability probabilities by (standard) regular reachability probabilities, using a construction in [**empty citation**].

Luko: I am not sure about either s_a or *alarm*, the first is shorter, but the second is a lot more descriptive

Example 6.

SJ: The definition below needs either that we use the horizon in the states explicitly, or that we use a depth funciton or so that gives us the distance of every state to a particular state.

SJ: Put a slightly longer proof in the appendix

Definition 14. *Given a set of states S, and an acyclic HMM $\mathcal{M} = \left(S' \subseteq^{fin} \mathbb{N} \times S, \iota, \mathbf{P}, Z, obs, r\right)$ with alarm state s_a, and an ignore and end observation as in Definition 13, we define the colored MDP $\left(\left(S', \iota, Act, \mathbf{P}'\right), C, c\right)$ with $Act = Z \setminus \{ignore\}$,*

$$\mathbf{P}'(s, z, q) := \begin{cases} \mathbf{P}'(s, q) & \textit{if } obs(q) = z \\ \sum_{q' \in S' \ \mid \ obs(q') \neq z} \mathbf{P}'(s, q') & \textit{if } q = \iota \\ 0 & \textit{otherwise} \end{cases}$$

$C = \{1, \ldots, h\}$ and c is given such that $c(\langle i, s\rangle) = i$.

Thus we transition normally to a state if the action and observation of the target state are equal, otherwise we set the transition probability to zero. All the remaining probability mass is redirected towards the initial state.

The idea behind this construction We fully remove the ignore action and ignore state, as they should never be taken[6].

Luko: Should we give a reason why this is correct? And not normalizing the probabilities?

SJ: What are the ingore action and ignore state?

The above construction allows for conditioning on a trace τ by constructing a policy σ that selects the ith observation of τ in the state with step i.

Definition 15 (Trace consistent policy). *Given two sets S and Z and an MDP $\mathcal{M}$ with states in $\mathbb{N} \times S$, and actions in $Z \cup \{end\}$. Together with a trace $\tau \in Z^\star$. A* trace consistent policy *is of the form*

$$\sigma_\tau(\langle i, s\rangle) = \begin{cases} \tau^{(i)} & \text{if } i \leq |\tau|, \\ end & \text{otherwise.} \end{cases}$$

Lemma 3.

$$\exists \sigma \in \Sigma_c . Pr_\sigma^{\mathcal{M}^C}(\Diamond T) \geq \lambda \quad \text{iff} \quad \exists \tau \in \mathcal{L}(\mathcal{M}). \sum_{\pi \in \Pi^{\mathcal{M}}} Pr^{\mathcal{M}}(\pi s_a \mid \tau \cdot z_{\mathsf{end}}) \geq \lambda,$$

In the ACTP HMM we only have the observations and conditional probability on a trace left. To incorporate this into the model we will use one novel idea. Instead of transitioning to a state and observing it, we will allow the model to choose what observation to observe. We thus move away from HMMs and into MDPs. This has two results. Firstly, instead of talking about sequences of observations, instead we will consider sequences of actions taken. Secondly, any trace can be transformed into a policy of the MDP. We will first discuss how to create this MDP. Afterward, we show how to ensure how, traces and policies can be made equivalent by coloring the MDP.

Corollary 2. *There exists a map $t(\sigma_\tau) = \tau$ which, given a trace consistent policy σ_τ finds its associated trace τ.*

However, when finding the maximal trace, we might not find a trace consistent policy. To fix this we color the MDP such that color-consistent policies coincide with trace consistent policies on our MDPs. As colors Given that all *end* states sinks, color consistent policies with the trace consistent policies.

Remark 3. All states gain all actions in $\mathbf{P}''$, we can later prune any actions for all states S^C of a color C when these actions only redirect to the initial state from S^C.

Now finding a trace with the conditional trace probability above λ_s coincides with finding a color consistent policy such that

$$Pr^\sigma(\Diamond \left\langle \begin{matrix} h+1, \\ alarm \end{matrix} \right\rangle) > \lambda_s$$

[6] The ignore state was included in ACTP to ensure we had no deadlock states.

3.4 Adapting to No False Alarms and Smaller Traces

- Add end transitions to non horizon states.
- We want to minimize alarm probability, not possible.
- Instead maximize safe probability.

4 Learning Monitors

- Use MAT
- Filtering as MQ [7]
- Monitor correctness as EQ
- If not correct returns counter example trace

Design choices

- Limit filtering to horizon
- Use Random walk with MQ as pre-EQ

5 Computational Complexity

Theorem 2. *Checking whether a monitor is correct given a unary encoded horizon is coNP-complete.*

In fact, we study in this section the dual to this problem, i.e., checking the existence of a counterexample. We call this problem *monitor co-verification*. For monitor co-verification, *membership* in NP follows as false alarms or a missed alarm (of length up to horizon) are witnesses. Verifying whether a trace is indeed a false alarm can be done in polynomial time by first checking whether the automaton accepts it and then computing the trace risk and comparing it to the thresholds. Each of these steps can be done in polynomial time [**empty citation**].

SJ: Maybe this result is so crucial that we should explain somewhere how this works in polynomial time. I guess it also follows from the construciton in section 3 if we restrict the monitor to just contain one trace?

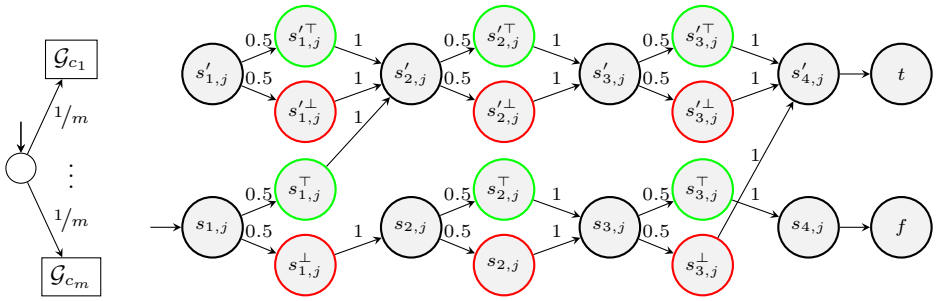
To establish NP-hardness, we consider the CTP problem from Definition 12. As a solution to the monitor co-verification problem suffices to solve the CTP problem (using a trivial monitor), this implies NP hardness of the former.

Lemma 4. *The CTP Decision Problem is NP-hard.*

The lemma holds even for $\lambda = 1$ and acyclic HMMs. We reduce the problem from 3CNF. We outline the construction below.

Consider a 3CNF formula $\varphi = \bigwedge c_1 \dots c_m$ over variables X, $|X| = n$, with clause $c_j = \ell_j^1 \vee \ell_j^2 \vee \ell_j^3$ and each literal $\ell_j^i \in \{x, \neg x \mid x \in X\}$. We transform this into a CTP instance with $\lambda = 1$ and an acyclic HMM $\mathcal{M}_\varphi$ with observations $Z = \{\#, \bot, \top\}$. The only state with positive risk is a dedicated state t with $r(t) = 1$. The crux of the construction is that there is a trace with risk 1 iff there is a satisfying assignment to φ. In the constructed HMM, there is a trace with risk 1 iff there is a trace where all corresponding paths end in state t.

SJ: make sure that corresponding paths are the right formal notion



(a) HMM. (b) Gadget demonstrated for $x_1 \vee \neg x_3$

Fig. 9: Illustrations for Lemma 4, with m clauses and variables x_1, x_2, x_3.

Before we give a formal definition of $\mathcal{M}_\varphi$, we give some intuition. We represent assignments $\alpha\colon X \to \{\bot, \top\}$ by traces through the HMM $\#\# \cdot \alpha(x_0) \cdot \# \cdot \alpha(x_1) \cdot \cdots \#$. We create gadgets for every clause. The gadgets are connected as in Figure 9a: That is, intuitively, $\mathcal{M}_\varphi$ randomly selects a clause c_j with probability $1/m$ and transitions into gadget $\mathcal{G}_j$ defined below. Next, we show that the gadget reaches positive risk in every gadget only if the corresponding clause is satisfied by the assignment.

The gadget $\mathcal{G}_j$ intuitively 'evaluates' c_j with respect to an assignment α, by starting in $s_{1,j}$ with a gadget for this clause, as exemplified in Fig. 9b. We enter the gadget with trace $\# \cdot \alpha(x_0) \cdot \# \cdot \alpha(x_1) \cdot \cdots \#$, i.e., the first observation has been matched by the initial state. A path (or its trace) through $\mathcal{G}_j$ 'reads' variable x_i in state $s_{i,j}$ and transitions to $s^{\top}_{i,j}$ or to $s^{\bot}_{i,j}$. However, only one of these paths corresponds to the trace. Thus, in every gadget, there is only one path that corresponds to a given trace and so they can be used interchangeably. State $s_{i,j}$, $s^{\top}_{i,j}$ and $s^{\bot}_{i,j}$ have observations $\#, \top, \bot$, respectively. We note that a path reaches $s^{\top}_{i,j}$ if $\alpha(x_i) = \top$ and (analogously for $\bot$). From there onwards, if the literal x_i (or $\neg x_i$) occurs in clause c_j, we transition from $s^{\top}_{i,j}$ (or $s^{\bot}_{i,j}$, resp.) to $s'_{i+1,j}$, otherwise, we transition to $s_{i+1,j}$. A path ends in t only via a state $s'_{i,j}$, which is only possible if the path visits a state corresponding to a literal in the clause. Thus, (conditionally) reaching state $s'_{i,j}$ with positive cconditional probability means that the partial assignment $\alpha(x_1), \ldots, \alpha(x_i)$ satisfies clause c_j, while reaching $s_{i,j}$ with positive conditional probability means that the clause is either unresolved or unsatisfied given the partial assignment.

Formally, we construct the HMM $\mathcal{M} = (S, \iota, \mathbf{P}, Z, obs, r)$ and set $\lambda = 1$:

- $S = \{s_{i,j}, s^{\bot}_{i,j}, s^{\top}_{i,j}, s'_{i,j}, s'^{\bot}_{i,j}, s'^{\top}_{i,j} \mid i \in \{1, \ldots, n+1\}, j \in \{1, \ldots, m\}\} \cup \{s_\iota\}$.
- $\mathbf{P}$ is given such that for all $i \in \{1, \ldots, n\}, j \in \{1, \ldots, m\}$:
  - $\mathbf{P}(s_\iota, s_{1,j}) = \frac{1}{m}$,
  - $\mathbf{P}(s_{i,j}, s^{\top}_{i,j}) = \mathbf{P}(s_{i,j}, s^{\bot}_{i,j}) = \frac{1}{2} = \mathbf{P}(s'_{i,j}, s'^{\top}_{i,j}) = \mathbf{P}(s'_{i,j}, s'^{\bot}_{i,j})$,
  - $\mathbf{P}(s^{\top}_{i,j}, s'_{i+1,j}) = [x_i = \ell^k_j$ for some k$]$, $\mathbf{P}(s^{\top}_{i,j}, s_{i+1,j}) = [x_i \neq \ell^k_j$ for all k$]$,
  - $\mathbf{P}(s^{\bot}_{i,j}, s'_{i+1,j}) = [\neg x_i = \ell^k_j$ for some k$]$, $\mathbf{P}(s^{\bot}_{i,j}, s_{i+1,j}) = [\neg x_i \neq \ell^k_j$ for all k$]$

  - $\mathbf{P}(s'_{n+1,j}, t) = 1 = \mathbf{P}(s_{n+1,j}, f)$
- $Z = \{\top, \bot, \#\}$ and $obs(s^{\top}_{i,j}) = (s'^{\top}_{i,j}) = \top$, $obs(s^{\bot}_{i,j}) = (s'^{\bot}_{i,j}) = \bot$, $obs(s_{i,j}) = obs(s'_{i,j}) = \# = obs(s_{\iota}) = obs(t) = obs(f)$.
- $r(t) = 1$ and $r(s) = 0$ for all $s \in S \setminus \{t\}$.

The construction runs in polynomial time. The formal proof of its correctness follows the explanation above precisely.

Approximability Approximating the verification of monitors is also hard, in the sense that it is hard to approximate the maximal risk that a monitor admits.

The maximal risk problem is an optimization problem

Definition 16 (CTP Optimization Problem).

Lemma 5. *The CTP Optimization Problem is APX-hard.*

We show this by means of an L-reduction from MAX-3SAT, which is an inapproximable and APX-hard problem [**empty citation**]. The construction follows along the reduction for the NP-hardness.

6 Experiments

We build our tool in the python programming language on top of the model checker Storm [6] and the POMDP model checker PAYNT [2]. This tool includes both the ToVer verifier algorithm and the learning algorithm employing ToVer for equality queries. inline]TODO: Write further description of the tool.

Luko: [

Benchmarks We evaluate both the ToVer verifier and the learning algorithm. We evaluate the ToVer verifier using the following questions:

$\mathbf{Q}^1_{VM}$ How does the runtime of ToVer verifier depend on its parameters?
$\mathbf{Q}^2_{VM}$ How much does the ToVer verification depend on PAYNT?
$\mathbf{Q}^3_{VM}$ Which PAYNT strategy is the most efficient in the ToVer benchmarks?

Luko: Maybe not a full question?

We also evaluate the learning algorithm by looking at the following questions:

$\mathbf{Q}^1_{L}$ Does ToVer learning guarantee a better bound then learning using a sampling EQ?
$\mathbf{Q}^2_{L}$ How do the ToVer learning monitors and sampling learning monitors compare in size?
$\mathbf{Q}^3_{L}$ Do monitors learned without a horizon limit in MQ generalize to outside the horizon?

Luko: Again, maybe less relevant?

7 Related

- LTLf
- Predictors or so by Ziemek et al.
- MDPs as Distribution transformers
- Computing conditional probabilities (About reachability conditionals, not ending conditionals)
- Fairness related to semi-consistence and consistence.

8 Conclusion and Future Work

- Long traces (complexity and practical implementation).
- Unbounded traces (It woudl be great if we can add why this is hard)

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